

RiskPulse: An Integrated Research Software Architecture for Multi-Hazard Evidence Governance, Remote Sensing, and Decision Support in Mountainous Terrain

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ABSTRACT:

We present RiskPulse, a software-verified research software architecture designed for multi-hazard disaster risk intelligence, evidence provenance tracking, and decision support in Himachal Pradesh, India. RiskPulse integrates local DEM terrain processing, Sentinel-2 spectral indices, multi-stream OSINT fusion, solver-neutral hydrodynamic adapters, and evidentiary briefing synthesis into a unified, traceable research environment. Crucially, RiskPulse enforces strict research-to-operational separation via a Controlled Promotion Gate. Within the audited software scope (348/348 tests passed GREEN, 0 Flutter analyzer

errors), deterministic computational pathways and software contracts are fully verified. We document the uncalibrated global reference status of the Caine landslide threshold ($I = 14.82 * D^{-0.39}$) and Rational flood model ($C=0.65$), explicitly distinguishing software verification from empirical scientific validation.

KEYWORDS: Geomatics; Natural Hazards; Remote Sensing; Disaster Risk Intelligence; Evidence Provenance; Himachal Pradesh.

1. INTRODUCTION & RESEARCH GAP

Disaster risk management in high-relief mountain catchments requires synthesizing disparate hydrometeorological observations, satellite remote sensing, and crowd-sourced incident reports. In high-relief regions such as the Indian Himalayas, data fragmentation and unverified model parameters often compromise operational warnings. RiskPulse addresses this architectural challenge by coupling multi-source evidence acquisition with deterministic calculation engines, explicit provenance tracking, and a Controlled Promotion Gate.

2. MATERIALS AND SYSTEM ARCHITECTURE

2.1 Copernicus GLO-30 DEM Processing & Hydrological Conditioning

Processing incorporates local 30m GeoTIFF rasters in EPSG:4326. D8 flow direction, Planchon-Darboux depression filling, Strahler/Shreve stream ordering, and watershed delineation operate deterministically in memory.

2.2 Sentinel-2 Multispectral Processing

Bands B2, B3, B4, and B8 (10m spatial resolution) apply ESA L2A reflectance scaling (0.0001) to compute NDVI and NDWI with zero-denominator safeguards.

2.3 OSINT Multi-Stream Evidence Fusion

Multi-stream incident reports apply spatial proximity (<100m) and Jaccard text similarity (>0.80) for evidence corroboration.

2.4 HydroAI Solver-Neutral Adapter

HydroAI couples terrain grids and hydrograph inputs to 2D hydrodynamic solvers. In the absence of native RasUnsteady64.exe, the engine executes via a software-verified simulated fallback mode (simulatedMock).

3. ANALYTICAL HAZARD MODELS & DECISION SUPPORT

3.1 Caine Landslide Threshold Model

Global power-law relationship $I = 14.82 * D^{-0.39}$ (alpha = 14.82, beta = -0.39) is preserved as an uncalibrated global

reference profile. Current project-defined calibration-readiness target is $n_caine_target \geq 10$ complete event pairs.

3.2 Rational Method Peak Discharge Model

Peak runoff $Q = (C * i * A) / 3.6$ applies an uncalibrated default runoff coefficient $C = 0.65$. Current project-defined hydrograph-readiness target is $n_hydro_target \geq 5$ continuous 15-minute event hydrographs.

3.3 SAR 2D Inundation Validation Framework

Dual-polarization Sentinel-1 GRD backscatter thresholding (-14dB) and spatial confusion matrix calculation are software-verified; real SAR validation rasters ($n_sar = 0$) remain uningested.

3.4 Relative Priority Decision Support Index

Analytical briefing synthesis computes a relative priority rank (0.0 ... 1.0) based on weighted evidence inputs. The priority index represents relative analytical priority, NOT an event occurrence probability.

4. SOFTWARE VERIFICATION & EVIDENCE RESULTS

All computational pathways passed 348/348 automated unit and integration tests. Five exploratory landslide events

(Kotropi, Nigulsari, Batseri, Boh, Summer Hill) demonstrate a descriptive mean observed-to-reference ratio of 1.340 against the global Caine curve.

5. SCIENTIFIC LIMITATIONS & FUTURE VALIDATION ROADMAP

Model parameters remain uncalibrated for regional terrain. Empirical hold-out validation datasets ($n_{\text{val}} = 0$, $n_{\text{hydro}} = 0$, $n_{\text{sar}} = 0$) remain incomplete. Native HEC-RAS execution and live GEE tile streaming remain unverified. Sample-size targets ($n_{\text{caine_target}} \geq 10$, $n_{\text{hydro_target}} \geq 5$, $n_{\text{val_target}} \geq 5$) are project-defined workflow readiness targets, NOT universal scientific minimums.

6. DISCUSSION, ETHICS & CONCLUSIONS

RiskPulse establishes a software-verified research software framework with transparent limitation disclosure and strict research-to-operational isolation enforced by a Controlled Promotion Gate.